

Propositions:

A number of words making a complete grammatical sentence having a sense and meaning and also meant an assertion in logic or mathematics is called a proposition. This assertion may be of two types - declarative and non-declarative. A proposition or statement is a declarative sentence that is either true or false.

For ex., "Three plus three equals six" and "Three plus three equals seven" are both statements, the first because it is true and the second because it is false. Similarly " $x+y > 1$ " is not a statement because for some values of  $x$  and  $y$  the sentence is true, whereas for others it is false.

Note:

- 1) The truth or falsity of a statement is called its truth value.
- 2) Since only two possible truth values are admitted this logic is sometimes called two-valued logic.
- 3) Questions, exclamations and commands are not propositions.
- 4) It is customary to represent simple statements by letters  $p, q, r, \dots$  known as proposition variables. Propositional variables can only assume two values, true or false.
- 5) There are also two propositional constants, T and F, that represent ~~true~~ true and false, respectively.

## Single Proposition:

A proposition consisting of only a single propositional variable or a single propositional constant is called an atomic (primary, primitive) proposition or simply proposition; that is they cannot be further subdivided.

## Compound Proposition:

A proposition obtained from the combinations of two or more propositions by means of logical operators or connectives of two or more propositions or by negating a single proposition is referred to molecular or composite or compound proposition.

## Connectives:

The words and phrases (or symbols) used to form compound propositions are called connectives. There are five basic connectives called Negation, Conjunction, Disjunction, Implication or Conditional and Equivalence or Biconditional. The following symbols are used to represent connectives.

## Compound Statement

— Negation  $\rightarrow \sim, \neg, \neg$  (not)

— Conjunction  $\rightarrow \wedge$  (and)

— Disjunction  $\rightarrow \vee$  (or)

— Implication or Conditional  $\rightarrow \Rightarrow, \rightarrow$  (if...then)

— Equivalence or Biconditional  $\rightarrow \Leftrightarrow, \leftrightarrow$  (if and only if)

| Symbolic form         | Negation of Symbolic form                      |
|-----------------------|------------------------------------------------|
| $\sim p$              | $\sim(\sim p) = p$                             |
| $p \wedge q$          | $(\sim p) \vee (\sim q)$                       |
| $p \vee q$            | $(\sim p) \wedge (\sim q)$                     |
| $p \Rightarrow q$     | $p \wedge (\sim q)$                            |
| $p \Leftrightarrow q$ | $[p \wedge (\sim q)] \vee [q \wedge (\sim p)]$ |

Negation: If  $p$  is any proposition, the negation of  $p$ , denoted <sup>③</sup> by  $\sim p$  or  $\neg p$  and read as not  $p$ , is a proposition which is false when  $p$  is true and when  $p$  is false. Consider the statement

$p$ : Paris is in France

$\sim p$ : Paris is not in France

Truth table:

| $p$ | $\sim p$ |
|-----|----------|
| T   | F        |
| F   | T        |

$p$ : All students are intelligent

$\sim p$ : Some students are not intelligent

$q$ : No student is intelligent

$\sim q$ : Some student are intelligent

Note:  $q$  is not the negation of  $p$

and  $p$  is not the negation of  $q$

Conjunction: If  $p$  and  $q$  are two statements, the conjunction of  $p$  and  $q$  is the compound statement denoted by  $p \wedge q$  and read as "p and q". The compound statement  $p \wedge q$  is true when both  $p$  and  $q$  are true, otherwise it is false.

Truth table:

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | F            |

Disjunction: If  $p$  and  $q$  are two statements, the disjunction of  $p$  and  $q$  is the compound statement denoted by  $p \vee q$  and read as "p or q". The statement  $p \vee q$  is true if at least one of  $p$  or  $q$  is true.

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| T   | T   | T          |
| T   | F   | T          |



Ex: Construct a truth table for each compound proposition (4)

i)  $p \wedge (\sim q \vee q)$

ii)  $\sim(p \vee q) \vee (\sim p \wedge \sim q)$



i)

| p | q | $\sim q$ | $\sim q \vee q$ | $p \wedge (\sim q \vee q)$ |
|---|---|----------|-----------------|----------------------------|
| T | T | F        | T               | T                          |
| T | F | T        | T               | T                          |
| F | T | F        | T               | F                          |
| F | F | T        | T               | F                          |

ii)

| p | q | $\sim p$ | $\sim q$ | $p \vee q$ | $\sim(p \vee q)$ | $\sim p \wedge \sim q$ | $\sim(p \vee q) \vee (\sim p \wedge \sim q)$ |
|---|---|----------|----------|------------|------------------|------------------------|----------------------------------------------|
| T | T | F        | F        | T          | F                | F                      | F                                            |
| T | F | F        | T        | T          | F                | F                      | F                                            |
| F | T | T        | F        | T          | F                | F                      | F                                            |
| F | F | T        | T        | F          | T                | T                      | T                                            |

### Algebra of Propositions:

Dual pairs:

- Replace all T by F
- Replace all F by T
- Replace all  $\wedge$  by  $\vee$
- Replace all  $\vee$  by  $\wedge$

### Laws of Algebra of Propositions:

i) Idempotent laws:

- a)  $p \vee p \equiv p$
- b)  $p \wedge p \equiv p$

2) Associative laws:

- a)  $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- b)  $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

3) Commutative laws:

$$a) p \vee q \equiv q \vee p$$

$$b) p \wedge q \equiv q \wedge p$$

4) Distributive laws:

$$a) p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$b) p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

5) Identity laws:

$$i) a) p \vee F \equiv p$$

$$b) p \wedge T \equiv p$$

$$ii) a) p \vee T \equiv T$$

$$b) p \wedge F \equiv F$$

6) Complement laws:

$$i) a) p \vee \sim p \equiv T$$

$$b) p \wedge \sim p \equiv F$$

$$ii) a) \sim T \equiv F$$

$$b) \sim F \equiv T$$

7) Involution laws:

$$\sim(\sim p) \equiv p$$

8) De Morgan's laws:

$$a) \sim(p \vee q) \equiv \sim p \wedge \sim q$$

$$b) \sim(p \wedge q) \equiv \sim p \vee \sim q$$

9) Absorption laws:

$$a) p \vee (p \wedge q) \equiv p$$

$$b) p \wedge (p \vee q) \equiv p$$

Ex: Use truth tables to prove the distributive law

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$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

| p | q | r | $q \wedge r$ | $p \vee (q \wedge r)$ | $p \vee q$ | $p \vee r$ | $(p \vee q) \wedge (p \vee r)$ |
|---|---|---|--------------|-----------------------|------------|------------|--------------------------------|
| T | T | T | T            | T                     | T          | T          | T                              |
| T | T | F | F            | T                     | T          | T          | T                              |
| T | F | T | F            | T                     | T          | T          | T                              |
| T | F | F | F            | T                     | T          | T          | T                              |
| F | T | T | T            | T                     | T          | T          | T                              |
| F | T | F | F            | F                     | T          | F          | F                              |
| F | F | T | F            | F                     | F          | T          | F                              |
| F | F | F | F            | F                     | F          | F          | F                              |

### Conditional Proposition:

If  $p$  and  $q$  are proposition, the compound proposition "if  $p$  then  $q$ " denoted by  $p \Rightarrow q$  is called a conditional proposition or implication and the connective is the conditional connective. The proposition  $p$  is called antecedent or hypothesis, and the proposition  $q$  is called the consequent or conclusion.

The only circumstances under which the implication  $p \Rightarrow q$  is false is when  $p$  is true and  $q$  is false.

Truth table:

| p | q | $p \Rightarrow q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

Ex: Calculate truth table for

(7)

i)  $p \vee \sim q \Rightarrow p$

ii)  $((\sim(p \wedge q) \vee r) \Rightarrow \sim p)$

i)

| p | q | $\sim q$ | $p \vee \sim q$ | $p \vee \sim q \Rightarrow p$ |
|---|---|----------|-----------------|-------------------------------|
| T | T | F        | T               | T                             |
| T | F | T        | T               | T                             |
| F | T | F        | F               | T                             |
| F | F | T        | T               | F                             |

ii)

| p | q | r | $p \wedge q$ | $\sim(p \wedge q)$ | $(\sim(p \wedge q) \vee r)$ | $((\sim(p \wedge q) \vee r) \Rightarrow \sim p)$ |
|---|---|---|--------------|--------------------|-----------------------------|--------------------------------------------------|
| T | T | T | T            | F                  | T                           | F                                                |
| T | T | F | T            | F                  | F                           | T                                                |
| T | F | T | F            | T                  | T                           | F                                                |
| T | F | F | F            | T                  | T                           | F                                                |
| F | T | T | F            | T                  | T                           | T                                                |
| F | T | F | F            | T                  | T                           | T                                                |
| F | F | T | F            | T                  | T                           | T                                                |
| F | F | F | F            | T                  | T                           | T                                                |

Ex: Use truth table to show that

$p \Rightarrow q \equiv \sim p \vee q$

→

| p | q | $p \Rightarrow q$ | $\sim p$ | $\sim p \vee q$ |
|---|---|-------------------|----------|-----------------|
| T | T | T                 | F        | T               |
| T | F | F                 | F        | F               |
| F | T | T                 | T        | T               |
| F | F | T                 | T        | T               |



## Converse, Contrapositive and Inverse :

If  $p \Rightarrow q$  is an implication, then the converse is  $q \Rightarrow p$   
the contrapositive is  $\sim q \Rightarrow \sim p$   
and the inverse is  $\sim p \Rightarrow \sim q$

Truth table :

| p | q | Conditional<br>$p \Rightarrow q$ | Converse<br>$q \Rightarrow p$ | Inverse<br>$\sim p \Rightarrow \sim q$ | Contrapositive<br>$\sim q \Rightarrow \sim p$ |
|---|---|----------------------------------|-------------------------------|----------------------------------------|-----------------------------------------------|
| T | T | T                                | T                             | T                                      | T                                             |
| T | F | F                                | T                             | T                                      | F                                             |
| F | T | T                                | F                             | F                                      | T                                             |
| F | F | T                                | T                             | T                                      | T                                             |

Note :

- 1) A conditional proposition and its converse or inverse are not logically equivalent.
- 2) A conditional proposition and its contrapositive are logically equivalent.
- 3) The importance of contrapositive derives from the fact that mathematical theorems in the form  $p \Rightarrow q$  can sometimes be proved more easily when restated in the form  $\sim q \Rightarrow \sim p$ .

Ex : Prove that if  $x^2$  is divisible by 4, then  $x$  is even.

→ Let  $p$  and  $q$  be the propositions such that  
 $p : x^2$  is divisible by 4,  
and  $q : x$  is even.

The implication is of the form  $p \Rightarrow q$ . The contrapositive is  $\sim q \Rightarrow \sim p$ , which states in words :

If  $x$  is odd, then  $x^2$  is not divisible by 4.

The proof of contrapositive is easy.

Since  $x$  is odd, we can write  $x = 2k+1$ , for some integer  $k$ . Hence  
 $x^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 4(k^2 + k + \frac{1}{4})$ . Since  $k^2 + k$  is an integer,  $k^2 + k + \frac{1}{4}$  is not an integer; therefore  $x^2$  is not divisible by 4.



## Biconditional Statement:

(9)

If  $p$  and  $q$  are statement ; then the compound statement  $p$  if and only if  $q$ , denoted by  $p \Leftrightarrow q$  is called a biconditional statement and the connective if and only if the biconditional connective.

The biconditional statement  $p \Leftrightarrow q$  can also be stated as " $p$  is a necessary and sufficient condition for  $q$ " or as " $p$  implies  $q$  and  $q$  implies  $p$ ".

In the truth table of  $p \Leftrightarrow q$ , it is true when both  $p$  and  $q$  are true or both  $p$  and  $q$  are false.

| $p$ | $q$ | $p \Leftrightarrow q$ |
|-----|-----|-----------------------|
| T   | T   | T                     |
| T   | F   | F                     |
| F   | T   | F                     |
| F   | F   | T                     |

Ex: Show that  $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$

| $p$ | $q$ | $p \Leftrightarrow q$ | $p \Rightarrow q$ | $q \Rightarrow p$ | $(p \Rightarrow q) \wedge (q \Rightarrow p)$ |
|-----|-----|-----------------------|-------------------|-------------------|----------------------------------------------|
| T   | T   | T                     | T                 | T                 | T                                            |
| T   | F   | F                     | F                 | T                 | F                                            |
| F   | T   | F                     | T                 | F                 | F                                            |
| F   | F   | T                     | T                 | T                 | T                                            |

Ex: Show that  $p \Leftrightarrow q \equiv (p \vee q) \Rightarrow (p \wedge q)$ , using  
a) truth table  
b) algebra of propositions.

| $p$ | $q$ | $p \Leftrightarrow q$ | $p \vee q$ | $p \wedge q$ | $p \vee q \Rightarrow p \wedge q$ |
|-----|-----|-----------------------|------------|--------------|-----------------------------------|
| T   | T   | T                     | T          | T            | T                                 |
| T   | F   | F                     | T          | F            | F                                 |
| F   | T   | F                     | T          | F            | F                                 |
| F   | F   | T                     | F          | F            | T                                 |

$$\begin{aligned}
b) \quad p \Leftrightarrow q &\equiv (p \Rightarrow q) \wedge (q \Rightarrow p) \\
&\equiv (\sim p \vee q) \wedge (\sim q \vee p) \\
&\equiv [(\sim p \vee q) \wedge \sim q] \vee [(\sim p \vee q) \wedge p] \quad \text{By distributive law} \\
&\equiv [\sim q \wedge (\sim p \vee q)] \vee [p \wedge (\sim p \vee q)] \quad \text{By commutative law} \\
&\equiv [(\sim q \wedge q) \vee (\sim q \wedge \sim p)] \vee [(p \wedge q) \vee (p \wedge \sim p)] \quad \text{By distributive law} \\
&\equiv [F \vee (\sim q \wedge \sim p)] \vee [(p \wedge q) \vee F] \quad \text{By complement law} \\
&\equiv (\sim q \wedge \sim p) \vee (p \wedge q) \quad \text{By Identity law} \\
&\equiv [\sim(p \vee q)] \vee (p \wedge q) \quad \text{By De Morgan's law} \\
&\equiv (p \vee q) \Rightarrow (p \wedge q)
\end{aligned}$$

### Negation of Compound Statements:

Negation of Conjunction: The negation of a conjunction  $p \wedge q$  is the disjunction of the negation of  $p$  and the negation of  $q$ . Equivalently, we write

$$\boxed{\sim(p \wedge q) \equiv \sim p \vee \sim q}$$

### Verification using truth table:

| $p$ | $q$ | $p \wedge q$ | $\sim(p \wedge q)$ | $\sim p$ | $\sim q$ | $\sim p \vee \sim q$ |
|-----|-----|--------------|--------------------|----------|----------|----------------------|
| T   | T   | T            | F                  | F        | F        | F                    |
| T   | F   | F            | T                  | F        | T        | T                    |
| F   | T   | F            | T                  | T        | F        | T                    |
| F   | F   | F            | T                  | T        | T        | T                    |

Negation of Disjunction : The negation of a disjunction  $p \vee q$  is the conjunction of the negation of  $p$  and the negation of  $q$ . Equivalently, we write

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

Verification using truth table :

| $p$ | $q$ | $\sim p$ | $\sim q$ | $\sim p \wedge \sim q$ | $p \vee q$ | $\sim(p \vee q)$ |
|-----|-----|----------|----------|------------------------|------------|------------------|
| T   | T   | F        | F        | F                      | T          | F                |
| T   | F   | F        | T        | F                      | T          | F                |
| F   | T   | T        | F        | F                      | T          | F                |
| F   | F   | T        | T        | T                      | F          | T                |

Negation of a Negation : A negation of negation of a statement is the statement itself. Equivalently, we write

$$\sim(\sim p) \equiv p$$

Verification using truth table :

| $p$ | $\sim p$ | $\sim(\sim p)$ |
|-----|----------|----------------|
| T   | F        | T              |
| F   | T        | F              |

Negation of Implication : If  $p$  and  $q$  are two statements, then

$$\sim(p \Rightarrow q) \equiv p \wedge \sim q$$

Verification using truth table :

| $p$ | $q$ | $p \Rightarrow q$ | $\sim(p \Rightarrow q)$ | $\sim q$ | $p \wedge \sim q$ |
|-----|-----|-------------------|-------------------------|----------|-------------------|
| T   | T   | T                 | F                       | F        | F                 |
| T   | F   | F                 | T                       | T        | T                 |
| F   | T   | T                 | F                       | F        | F                 |
| F   | F   | T                 | F                       | T        | F                 |

Negation of Biconditional : If  $p$  and  $q$  are two statements, then (12)

$$\boxed{\sim(p \Leftrightarrow q) \equiv p \Leftrightarrow \sim q \equiv \sim p \Leftrightarrow q}$$

Verification using truth table :

| $p$ | $q$ | $p \Leftrightarrow q$ | $\sim(p \Leftrightarrow q)$ | $\sim p$ | $\sim p \Leftrightarrow q$ | $\sim q$ | $p \Leftrightarrow \sim q$ |
|-----|-----|-----------------------|-----------------------------|----------|----------------------------|----------|----------------------------|
| T   | T   | T                     | F                           | F        | F                          | F        | F                          |
| T   | F   | F                     | T                           | F        | T                          | T        | T                          |
| F   | T   | F                     | T                           | T        | T                          | F        | T                          |
| F   | F   | T                     | F                           | T        | F                          | T        | F                          |

Note :

$$\begin{aligned} \sim(p \Leftrightarrow q) &\equiv \sim[(p \Rightarrow q) \wedge (q \Rightarrow p)] \\ &\equiv \sim[(\sim p \vee q) \wedge (\sim q \vee p)] \\ &\equiv (p \wedge \sim q) \vee (q \wedge \sim p) \end{aligned}$$

Derived Connectives :

⇒ NAND : It means negation of conjunction of two statements.

Assume  $p$  and  $q$  be two propositions. NAND of  $p$  and  $q$  is a proposition which is false when both  $p$  and  $q$  are true otherwise true. It is denoted by  $p \uparrow q$ .

$$\boxed{p \uparrow q \equiv \sim(p \wedge q)}$$

| $p$ | $q$ | $p \uparrow q$ |
|-----|-----|----------------|
| T   | T   | F              |
| T   | F   | T              |
| F   | T   | T              |
| F   | F   | T              |



# Tautologies and Contradictions:

(14)

Tautology: A compound proposition that is always true.

Contradiction: A compound proposition that is always false.

Contingency: A proposition that is neither a tautology nor a contradiction.

Ex: Prove that the following propositions are tautology

a)  $p \vee \neg p$

b)  $\neg(p \wedge q) \vee q$

c)  $p \Rightarrow (p \vee q)$

| p | $\neg p$ | $p \vee \neg p$ |
|---|----------|-----------------|
| T | F        | T               |
| F | T        | T               |

It can be seen from the last col<sup>n</sup> of the truth table that the expression is true for all possible assignments of p and q. Hence proposition is a tautology.

b)

| p | q | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg(p \wedge q) \vee q$ |
|---|---|--------------|--------------------|---------------------------|
| T | T | T            | F                  | T                         |
| T | F | F            | T                  | T                         |
| F | T | F            | T                  | T                         |
| F | F | F            | T                  | T                         |

Same as (a)

c)

| p | q | $p \vee q$ | $p \Rightarrow (p \vee q)$ |
|---|---|------------|----------------------------|
| T | T | T          | T                          |
| T | F | T          | T                          |
| F | T | T          | T                          |
| F | F | F          | T                          |

Same as (a)

$$\begin{aligned}
 & \neg (p \vee q) \vee (r \Leftrightarrow s) \equiv (p \Rightarrow q \vee r) \vee ((\neg r \vee s) \wedge (r \vee \neg s)) \\
 & \equiv (\neg p \vee \neg q \vee r) \vee ((\neg r \vee s) \wedge (r \vee \neg s)) \\
 & [p \Rightarrow q \wedge r \equiv \neg p \vee (q \wedge r)]
 \end{aligned}$$

### Well Formed Formula:

A statement formula is an expression which is a string consisting of variables, parentheses and connective symbols. A grammatically ~~correct~~ correct expression is called a well-formed formula, which is abbreviated wff.

A wff can be generated by the following rules:

- 1) All variables and constants are well-formed formulae.
- 2) If  $P$  is a well-formed formula, then  $\neg P$  is also a well-formed formula.
- 3) If  $P$  and  $Q$  are well-formed formulae, then  $P \wedge Q$ ,  $P \vee Q$ ,  $P \Rightarrow Q$  and  $P \Leftrightarrow Q$  are all well-formed formulae.
- 4) A statement formula consists of variables, parentheses and connectives is recursively well-formed formula iff it can be obtained by finitely applying the above rules.
- 5) Nothing else is a well-formed formula.

For ex.,  $\neg (P \vee Q)$  and  $(P \Rightarrow (P \wedge Q))$  are well-formed formulae  
 whereas  $\neg Q$  and  $(P \wedge Q) \Rightarrow (\neg P)$  are not.

## Normal Forms:

(17)

By comparing truth tables, one determine whether two logical expressions  $P$  and  $Q$  are equivalent. But the process is very tedious when the number of variables increases. A better method is to transform the expressions  $P$  and  $Q$  to some standard forms of expressions  $P'$  and  $Q'$  such that a simple comparison of  $P'$  and  $Q'$  shows whether  $P \equiv Q$ . The standard forms are called normal forms or canonical forms.

There are two types of normal forms:

- 1) Disjunctive normal forms (dnf) and
- 2) Conjunctive normal forms (cnf)

Note: It will be convenient to use the words product and sum in place of the logical connectives conjunction and disjunction.

### Disjunctive normal form (dnf):

In a logical expression, a product of the variables and their negations is called an elementary product. For ex,  $p \wedge q$ ,  $\neg p \wedge \neg q$ ,  $\neg p \wedge q$  are elementary products.

A sum of the variables and their negations is called an elementary sum. For ex,  $p \vee q$ ,  $p \vee \neg q$ ,  $\neg p \vee \neg q$  are elementary sums.

The elementary sums or products satisfy the following properties.

- 1) An elementary product is identically false if and only if it contains at least one pair of factors in which one is negation of the other.
- 2) An elementary sum is identically true if and only if it contains at least one pair of factors in which one is negation of the other.

A logical expression is said to be in disjunctive normal form (18) if it is the sum of elementary products.

For ex,  $p \vee (q \wedge r)$  and  $p \vee (\neg q \wedge r)$  are in dnf,

$p \wedge (q \vee r)$  is not in dnf.

Procedure to obtain a dnf of a given logical expression:

Step 1: Remove all  $\Rightarrow$  and  $\Leftrightarrow$  by an equivalent expression containing the connectives  $\wedge, \vee, \neg$  only.

Step 2: Eliminate  $\sim$  before sums and products by using the double negation or by using De Morgan's laws.

Step 3: Apply the distributive law until a sum of elementary products is obtained.

Ex: Obtain the disjunctive normal forms of the followings:

a)  $p \wedge (p \Rightarrow q)$

b)  $p \vee (\neg p \Rightarrow (q \vee (q \Rightarrow \neg r)))$

c)  $p \Rightarrow ((p \Rightarrow q) \wedge \sim(\neg q \vee \neg p))$

a)  $p \wedge (p \Rightarrow q) \equiv p \wedge (\neg p \vee q) \equiv (p \wedge \neg p) \vee (p \wedge q)$

which is the required dnf.

b)  $p \vee (\neg p \Rightarrow (q \vee (q \Rightarrow \neg r))) \equiv p \vee (\neg p \Rightarrow (q \vee (\neg q \vee \neg r)))$   
 $\equiv p \vee (p \vee (q \vee (\neg q \vee \neg r)))$

$\equiv p \vee p \vee q \vee \neg q \vee \neg r$

$\equiv p \vee q \vee \neg q \vee \neg r$

which is the required dnf.



$$\begin{aligned}
 c) \quad p \Rightarrow ((p \Rightarrow q) \wedge \sim (\sim q \vee \sim p)) &\equiv \sim p \vee ((p \Rightarrow q) \wedge \sim (\sim q \vee \sim p)) \\
 &\equiv \sim p \vee ((\sim p \vee q) \wedge \sim (\sim q \vee \sim p)) \\
 &\equiv \sim p \vee ((\sim p \vee q) \wedge (q \wedge p)) \\
 &\equiv \sim p \vee [(\sim p \wedge (q \wedge p)) \vee (q \wedge (q \wedge p))] \\
 &\equiv \sim p \vee [(\sim p \wedge q) \wedge p] \vee [q \wedge q \wedge p] \\
 &\equiv \sim p \vee F \vee (p \wedge q) \\
 &\equiv \sim p \vee (p \wedge q)
 \end{aligned}$$

which is the required def.

note:

def of a given logical expression is not unique. For ex, consider  $p \vee (q \wedge r)$ . This is already in def.

$$\begin{aligned}
 \text{We can write } p \vee (q \wedge r) &\equiv (p \vee q) \wedge (p \vee r) \\
 &\equiv (p \wedge p) \vee (p \wedge q) \vee (p \wedge r) \vee (q \wedge r)
 \end{aligned}$$

and this is another equivalent normal form.

Conjunctive Normal Form (cnf):

A logical expression is said to be in cnf if it consists of a product of elementary sum.

Ex: Obtain a cnf of the following

$$\begin{aligned}
 a) \quad p \wedge (p \Rightarrow q) \\
 b) \quad [q \vee (p \wedge r)] \wedge \sim [(p \vee r) \wedge q]
 \end{aligned}$$

$\rightarrow$  a)  $p \wedge (p \Rightarrow q) \equiv p \wedge (\sim p \vee q)$  which is the required cnf.

$$\begin{aligned}
 b) \quad [q \vee (p \wedge r)] \wedge \sim [(p \vee r) \wedge q] &\equiv [q \vee (p \wedge r)] \wedge [\sim (p \vee r) \vee \sim q] \\
 &\equiv [\bar{q} \vee (p \wedge r)] \wedge [(\sim p \wedge \sim r) \vee \sim q] \\
 &\equiv (q \vee p) \wedge (q \vee r) \wedge (\sim p \vee \sim q) \wedge (\sim r \vee \sim q) \\
 &\text{which is the required cnf.}
 \end{aligned}$$

## Principal Disjunctive Normal Form (PDNF)

Let  $p$  and  $q$  be two  $0/1$ -valued variables, then  $p \wedge q$ ,  $p \wedge \neg q$ ,  $\neg p \wedge q$  and  $\neg p \wedge \neg q$  are called minterms of  $p$  and  $q$ . It may be noted that none of the minterms should contain both a variable and its negation. For given two variables there are  $2^2$  minterms. The number of minterms in  $n$  variables is  $2^n$ .

Principal Disjunctive Normal Form of a given formula can be defined as an equivalent formula consisting of disjunctives of minterms only. This is also called the sum of products canonical form. The process for obtaining principal disjunctive normal form is discussed below.

### (I) By truth table:

- 1) Construct a truth table of the given compound propositions.
- 2) For every truth value  $T$  of the given proposition, select the minterm, which also has the value  $T$  for the same combination of the truth value of the  $0/1$ -valued variables.
- 3) The disjunctive of the minterms selected in 2 is the required principal disjunctive normal form.

### (II) Without constructing truth table:

- 1) Obtain a disjunctive normal form.
- 2) Group elementary products which are contradictions (such as  $p \wedge \neg p$ )
- 3) If  $p_i$  and  $\neg p_i$  are missing in an elementary product  $\alpha$ , replace  $\alpha$  by  $(\alpha \wedge p_i) \vee (\alpha \wedge \neg p_i)$ .

- 4) Repeat step 3 until all elementary products are reduced to sum of minterms. Identical minterms appearing in the disjunction are deleted.

The advantages of obtaining principal disjunctive normal form are:

- i) The principal disjunctive normal form of a given formula is unique.
- ii) Two formulas are equivalent if and only if their principal disjunctive normal forms coincide.
- iii) If the given compound proposition is a tautology, then its principal disjunctive normal form will contain all possible minterms of its components.

Ex: Obtain the principal disjunctive normal form of

a)  $p \Rightarrow q$

b)  $q \vee (p \vee \sim q)$

c)  $\sim p \vee q$

d)  $(p \wedge \sim q \wedge \sim r) \vee (q \wedge r)$

- (a) and (b) using the truth table,  
(c) and (d) without using the truth table.

a)  $\rightarrow$

| p | q | $p \Rightarrow q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

Truth table of  $p \Rightarrow q$

Thus principal disjunctive normal form of  $p \Rightarrow q$  is  $(p \wedge q) \vee (\sim p \wedge q) \vee (\sim p \wedge \sim q)$ .



## Principal Conjunctive Normal forms (pcnf):

(23)

The dual of a minterm is called a maxterm. For a given number of variables, the maxterm consists of disjunctions in which each variable or its negation, but not both, appears only once. For ex., for two variables  $p$  and  $q$ , there are maxterms given by

$$p \vee q, p \vee \sim q, \sim p \vee q, \sim p \vee \sim q.$$

Principal conjunctive normal form of a given formula can be defined as an equivalent formula consisting of conjunctive of maxterm only. This is also called the product of sums canonical form. The process for obtaining principal conjunctive normal form is similar to the one followed for principal disjunctive normal form.

For obtaining principal conjunctive normal form of  $\alpha$ , one can also construct the principal disjunctive normal form of  $\sim \alpha$  and apply negation.

The advantages of obtaining principal conjunctive normal form are:

i) The principal conjunctive normal form is unique.

ii) Every compound proposition, which is not a tautology, has an equivalent principal conjunctive normal form.

iii) If the given compound proposition is a contradiction, then its principal conjunctive normal form will contain all possible maxterms of its components.



Ex: Obtain the principal conjunctive normal form of

a)  $p \wedge q$  using truth table

b)  $(\sim p \Rightarrow r) \wedge (q \Leftrightarrow p)$  without using truth table

c)  $A = (p \wedge q) \vee (\sim p \wedge q) \vee (q \wedge r)$  by constructing principal disjunctive normal form.

a) Truth table of  $p \wedge q$ :

| p | q | $p \wedge q$ |
|---|---|--------------|
| T | T | T            |
| T | F | F            |
| F | T | F            |
| F | F | F            |

The required principal conjunctive normal form of  $p \wedge q$  is

$$(\sim p \vee \sim q) \wedge (p \vee \sim q) \wedge (p \vee q)$$

$$\text{b) } (\sim p \Rightarrow r) \wedge (q \Leftrightarrow p) \equiv (p \vee r) \wedge [(q \Rightarrow p) \wedge (p \Rightarrow q)]$$

$$\equiv (p \vee r) \wedge [(q \vee \sim p) \wedge (\sim p \vee q)]$$

$$\equiv [(p \vee r) \vee (q \vee \sim q)] \wedge [(q \vee \sim p) \vee (\sim p \vee q)]$$

$$\wedge [(p \vee q) \vee (r \vee \sim r)]$$

$$\equiv (p \vee r \vee q) \wedge (p \vee r \vee \sim q) \wedge (q \vee p \vee r)$$

$$\wedge (q \vee \sim p \vee \sim r) \wedge (\sim p \vee q \vee r)$$

$$\wedge (\sim p \vee q \vee \sim r)$$

$$\equiv (p \vee r \vee q) \wedge (p \vee r \vee \sim q) \wedge (q \vee p \vee r)$$

$$\wedge (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r)$$

which is the required principal conjunctive normal form,